```
# Task1
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

subjects = ["Math", "Science", "English"]
hours = [2, 3, 1]

df = pd.DataFrame({"Subject": subjects, "Hours": hours})

total = np.sum(df["Hours"])
avg = np.mean(df["Hours"])

print("Total Study Time:", total)
print("Average Study Time:", avg)

plt.bar(df["Subject"], df["Hours"])
plt.title("Study Time")
plt.show()

# Task2
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

food = ["Rice", "Milk", "Egg"]
calories = [200, 150, 100]

df = pd.DataFrame({"Food": food, "Calories": calories})
total = np.sum(df["Calories"])
print("Total Calories:", total)

plt.pie(df["Calories"], labels=df["Food"], autopct="%1.1f%%")
plt.title("Calorie Intake")
plt.show()
```

 # Task3

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

products = ["Pen", "Book", "Bag"]
quantity = [50, 30, 20]

df = pd.DataFrame({"Product": products, "Quantity": quantity})

total = np.sum(df["Quantity"])
print("Total Stock:", total)

plt.bar(df["Product"], df["Quantity"])
plt.title("Stock Levels")
plt.show()
```

 # Task4

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

numbers = [10, 20, 30, 40, 50]

df = pd.DataFrame({"Numbers": numbers})

print("Max:", np.max(df["Numbers"]))
print("Min:", np.min(df["Numbers"]))
print("Average:", np.mean(df["Numbers"]))

plt.plot(df["Numbers"])
plt.title("Number Comparison")
plt.show()
```

 # Task5

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

players = ["A", "B", "C"]
scores = [80, 60, 90]

df = pd.DataFrame({"Player": players, "Score": scores})

print("Highest Score:", np.max(df["Score"]))

plt.bar(df["Player"], df["Score"])
plt.title("Game Scores")
plt.show()
```

 # Task6

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

hours = ["9AM", "10AM", "11AM", "12PM"]
vehicles = [50, 80, 120, 90]

df = pd.DataFrame({"Time": hours, "Vehicles": vehicles})

peak = df.iloc[np.argmax(df["Vehicles"])]
print("Peak Time:", peak["Time"])

plt.plot(df["Time"], df["Vehicles"])
plt.title("Traffic Analysis")
plt.show()
```

 # Task7

TP = 50, FP = 5, FN = 10, TN = 35

Total = 50 + 10 + 5 + 35 = 100

Accuracy = (50 + 35) / 100 = 0.85

Precision = 50 / (50 + 5) = 0.909

Recall = 50 / (50 + 10) = 0.8333

F1 Score = 2 × (0.909 × 0.8333) / (0.909 + 0.8333)

F1 Score = 2 × 0.7574 / 1.7423

F1 Score = 0.8694

Total = 40 + 5 + 5 + 10 + 30 + 10 + 5 + 5 + 40 = 150

Accuracy = (40 + 30 + 40) / 150 = 0.7333

Precision (A) = 40 / (40 + 10 + 5) = 0.7272

Recall (A) = 40 / (40 + 5 + 5) = 0.8

F1 Score (A) = 2 × (0.7272 × 0.8) / (0.7272 + 0.8)

F1 Score (A) = 2 × 0.5817 / 1.5272

F1 Score (A) = 0.7618

Precision (B) = 30 / (5 + 30 + 5) = 0.75

Recall (B) = 30 / (10 + 30 + 10) = 0.6

F1 Score (B) = 2 × (0.75 × 0.6) / (0.75 + 0.6)

F1 Score (B) = 2 × 0.45 / 1.35

F1 Score (B) = 0.6666

Precision (C) = 40 / (5 + 10 + 40) = 0.7272

Recall (C) = 40 / (5 + 5 + 40) = 0.8

F1 Score (C) = 2 × (0.7272 × 0.8) / (0.7272 + 0.8)

F1 Score (C) = 2 × 0.5817 / 1.5272

F1 Score (C) = 0.7618

TP = 20, FP = 10, FN = 80, TN = 890

Total = 20 + 80 + 10 + 890 = 1000

Accuracy = (20 + 890) / 1000 = 0.91

Precision = 20 / (20 + 10) = 0.6666

Recall = 20 / (20 + 80) = 0.2

F1 Score = 2 × (0.6666 × 0.2) / (0.6666 + 0.2)

F1 Score = 2 × 0.1333 / 0.8666

F1 Score = 0.3076

TP = 45, FP = 15, FN = 5, TN = 35

Total = 45 + 5 + 15 + 35 = 100

Accuracy = (45 + 35) / 100 = 0.8

Precision = 45 / (45 + 15) = 0.75

Recall = 45 / (45 + 5) = 0.9

F1 Score = 2 × (0.75 × 0.9) / (0.75 + 0.9)

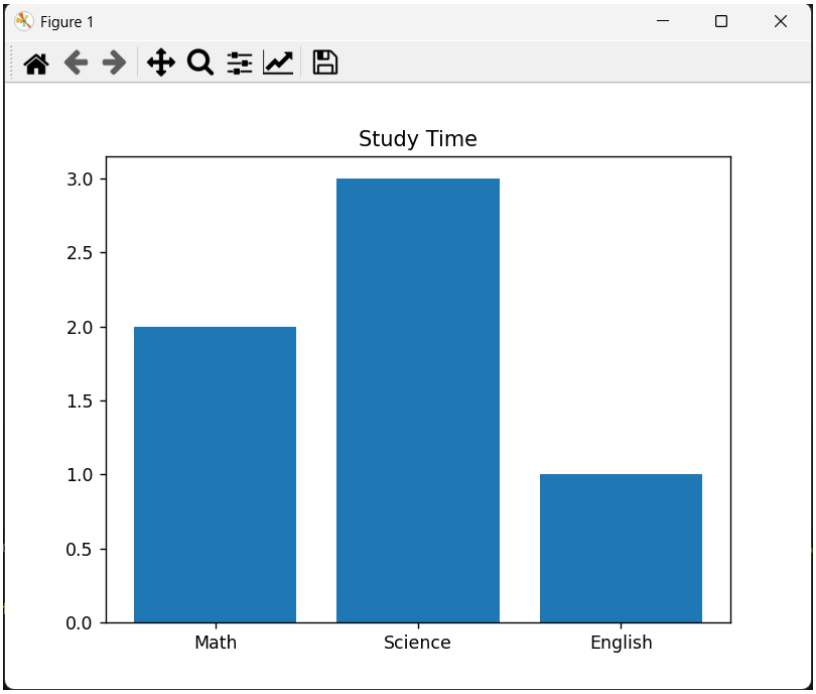
F1 Score = 2 × 0.675 / 1.65

F1 Score = 0.8181

"""

<u>Outputs:</u>

```
(.venv) PS D:\Internship\Day40> python task.py
Total Study Time: 6
Average Study Time: 2.0
Total Calories: 450
Total Stock: 100
Max: 50
Min: 10
Average: 30.0
Highest Score: 90
Peak Time: 11AM
```



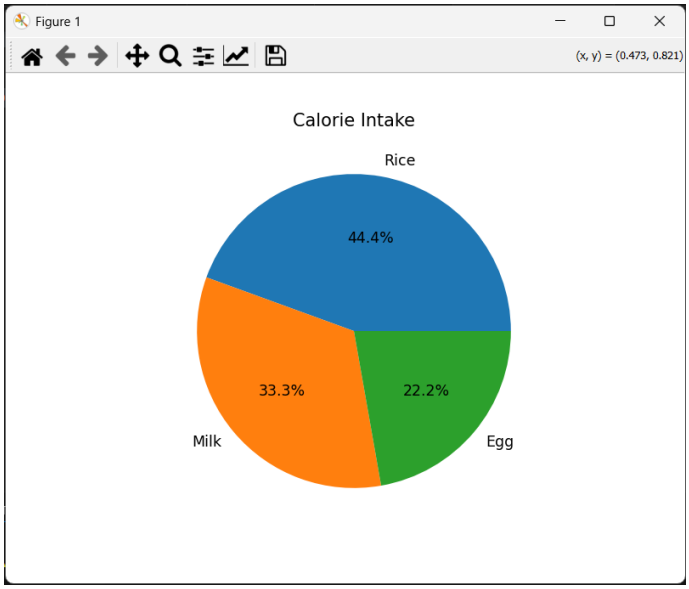


Figure 1
(x, y) = (0.473, 0.821)
Calorie Intake
Rice
44.4%
33.3%
22.2%
Milk
Egg

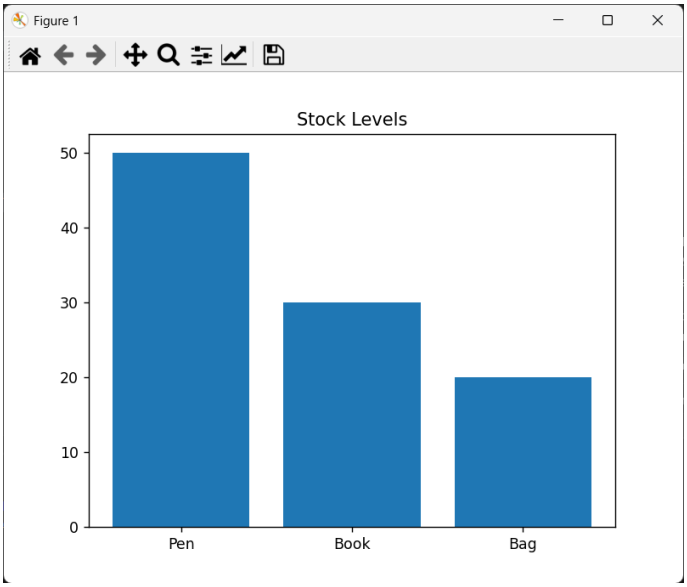


Figure 1
Stock Levels
50
40
30
20
10
0
Pen
Book
Bag

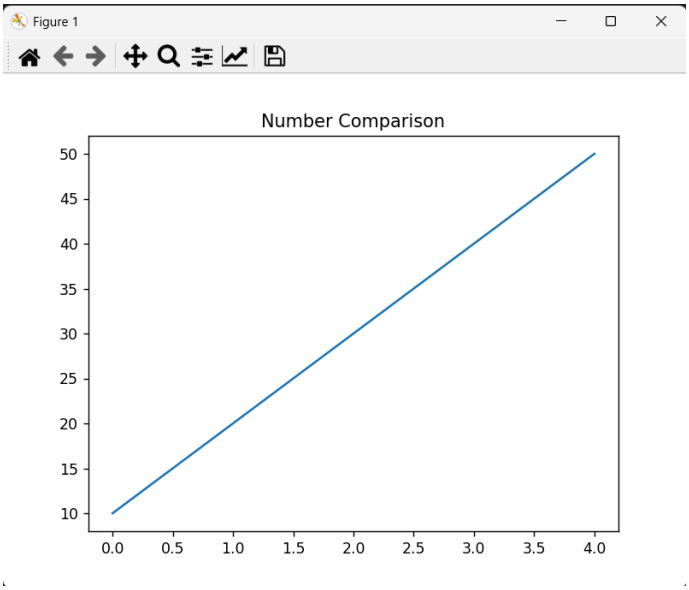


Figure 1
Number Comparison
50
45
40
35
30
25
20
15
10
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

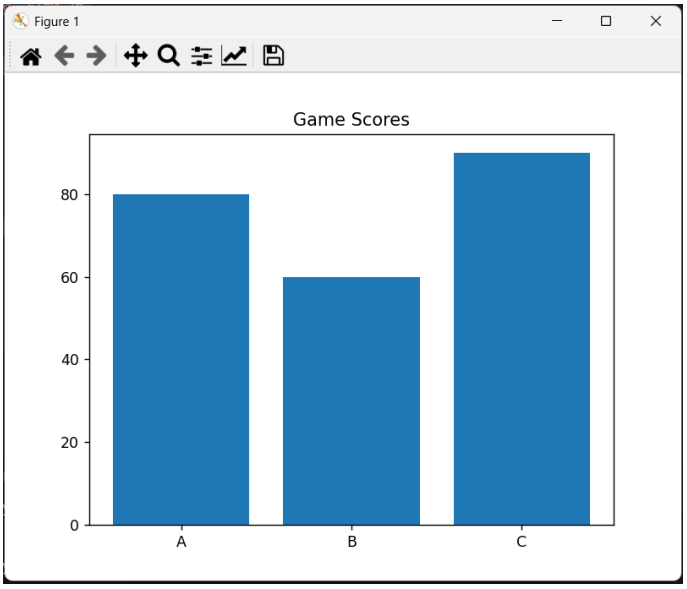


Figure 1
Game Scores
80
60
40
20
0
A
B
C

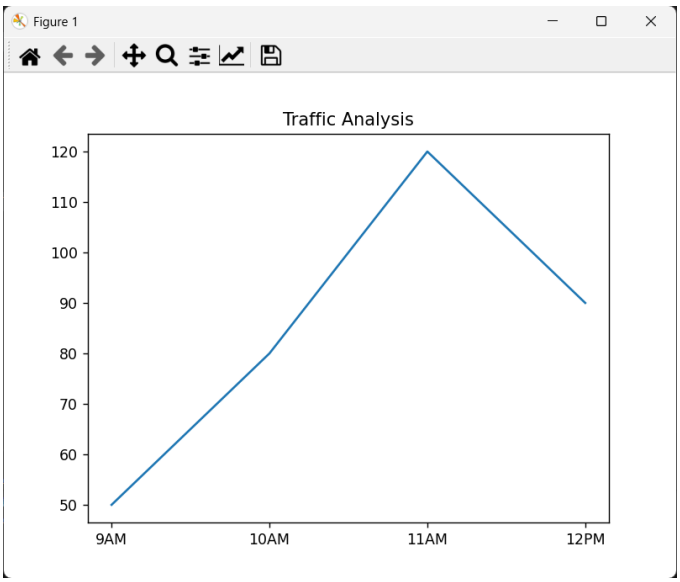


Figure 1
Traffic Analysis
120
110
100
90
80
70
60
50
9AM
10AM
11AM
12PM